

Cross-dataset enrichment of hypocotyl- and vascular-associated programs

Quantitative test across three independent spaceflight and radiation transcriptomic studies (OSD-120 root, OSD-658 whole seedling, OSD-678 leaf; 6 contrasts; 22,178–24,725 genes per contrast).

Headline

All **7 vascular-associated programs are significantly enriched** (BH-FDR ≤ 0.04 , all down-regulated). **None of the 3 hypocotyl-associated programs are significantly enriched** (BH-FDR ≥ 0.34) when tested across all contrasts simultaneously. The vascular composite (457 genes, union of 6 vascular cell-type panels) gives the strongest combined signal ($Z = -6.23$, BH-FDR = 4.55×10^{-9}) with 6/6 contrasts directionally consistent.

Significant programs (BH-FDR < 0.05)

Program	Atlas	n_panel	median log2FC	sign concordance	BH-FDR
vascular_composite	composite	457	-0.264	6/6	4.6e-09
Xylem_root	Han	100	-0.403	6/6	1.4e-08
Vasculature_shoot	Han	100	-0.343	5/6	1.0e-06
Liew_xylem	Liew	100	-0.242	5/6	6.3e-06
Procambium_root	Han	67	-0.220	5/6	9.4e-04
Phloem_root	Han	100	-0.113	5/6	0.005
Liew_provasculature	Liew	100	-0.112	4/6	0.039

Non-significant programs

Program	n_panel	median log2FC	sign concordance	BH-FDR
E.hypocotyl_epidermis_shoot	100	-0.045	4/6	0.335
hypocotyl_axis_composite	160	-0.065	4/6	0.681
Cortex_hypocotyl_shoot	67	-0.034	4/6	0.941

Why the asymmetry

The hypocotyl_axis_composite flips direction across contexts: it is strongly positive in OSD-120 root-tip dark (+0.64) and OSD-678 leaf-light (+0.52), strongly negative in OSD-678 leaf-dark (−0.68), and near-zero in both OSD-658 GCR contrasts (−0.06, −0.07). This pattern is consistent with the Stage E model that hypocotyl elongation programs are context- and tissue-specific. Vascular programs, by contrast, are monotonically suppressed across all 6 spaceflight and GCR contrasts.

Method

Per-contrast permutation against random gene sets of matched size (1,000 perms, sampled without replacement from the contrast's measured-gene universe). Stouffer combination across the 6 contrasts with $\sqrt{n}$ weighting,

sign-aware. BH-FDR across the 10 program-level p-values. Null calibration verified empirically at n=500 (median p = 0.505, type-I error 3.8% at $\alpha=0.05$).

Caveats

- 1. Scope.** 3 OSD studies only. Maffei2022 was excluded after coverage QC showed 0–4 gene overlap between its published 194-gene ROS/hormone/clock panel and the cell-type marker panels tested here.
- 2. Test conservatism.** Mean-extreme permutation is more conservative than rank-shift (Wilcoxon) for panels with small but consistent log2FC. Stage C reported a Wilcoxon FDR of 1.2×10^{-3} for the hypocotyl composite at OSD-658_80cGy; the matched permutation p here is 0.07. The headline is therefore a conservative test of cross-dataset enrichment; the hypocotyl signal still exists in *individual* contrasts (notably OSD-120 root-dark and OSD-678 leaf-dark) but does not accumulate across the full 6-contrast pool.
- 3. Sign-concordance $\neq$ significance.** 4/6 concordance is observed for both significant vascular programs (Liew_provasculature) and non-significant hypocotyl programs. What separates them is per-contrast effect magnitude, not direction agreement alone.
- 4. Cross-study heterogeneity.** OSD-658 is whole-seedling GCR irradiation; OSD-120 is root-tip microgravity; OSD-678 is leaf microgravity with dark/light conditions. Effect sizes differ by an order of magnitude between studies, which contributes to the per-contrast variability for hypocotyl programs.

Files

tables/cross_dataset_enrichment_meta.csv (wide, 10 rows): program-level meta-statistics including BH-FDR and sign concordance.

tables/cross_dataset_enrichment_meta_long.csv (long, 60 rows): per-(program $\times$ contrast) effect sizes and permutation p-values.

Per-contrast detail (60 tests)

Each row is one (program × contrast) cell from the long table. mean_log2FC is the panel mean; perm_p is the two-sided permutation p (1,000 perms).

vascular_composite (composite, n_panel=457) — Z = -6.23, BH-FDR = 4.6e-09, sign concordance = 6/6 **[SIGNIFICANT]**

Study	Tissue	Contrast	n_measured	mean log2FC	perm_p
OSD-120	root_tip	FLT_vs_GC_Dark	451	-0.426	0.0010
OSD-120	root_tip	FLT_vs_GC_Light	451	-0.262	0.0010
OSD-658	whole_seedling	GCR_40cGy_vs_NoRad	447	-0.267	0.0010
OSD-658	whole_seedling	GCR_80cGy_vs_NoRad	447	-0.126	0.0010
OSD-678	leaf	FLT_vs_GC_Dark	453	-0.304	0.8801
OSD-678	leaf	FLT_vs_GC_Light	453	-0.038	0.0490

Xylem_root (Han, n_panel=100) — Z = -5.94, BH-FDR = 1.4e-08, sign concordance = 6/6 **[SIGNIFICANT]**

Study	Tissue	Contrast	n_measured	mean log2FC	perm_p
OSD-120	root_tip	FLT_vs_GC_Dark	100	-0.904	0.0010
OSD-120	root_tip	FLT_vs_GC_Light	100	-0.058	0.8202
OSD-658	whole_seedling	GCR_40cGy_vs_NoRad	100	-0.281	0.0010
OSD-658	whole_seedling	GCR_80cGy_vs_NoRad	100	-0.166	0.0010
OSD-678	leaf	FLT_vs_GC_Dark	100	-0.524	0.2448
OSD-678	leaf	FLT_vs_GC_Light	100	-1.087	0.0010

Vasculature_shoot (Han, n_panel=100) — Z = -5.12, BH-FDR = 1.0e-06, sign concordance = 5/6 **[SIGNIFICANT]**

Study	Tissue	Contrast	n_measured	mean log2FC	perm_p
OSD-120	root_tip	FLT_vs_GC_Dark	98	-0.365	0.0010
OSD-120	root_tip	FLT_vs_GC_Light	98	-0.681	0.0010
OSD-658	whole_seedling	GCR_40cGy_vs_NoRad	97	-0.359	0.0010
OSD-658	whole_seedling	GCR_80cGy_vs_NoRad	97	-0.142	0.0050
OSD-678	leaf	FLT_vs_GC_Dark	99	-0.327	0.9061
OSD-678	leaf	FLT_vs_GC_Light	99	+0.212	0.8222

Liew_xylem (Liew, n_panel=100) — Z = -4.71, BH-FDR = 6.3e-06, sign concordance = 5/6 **[SIGNIFICANT]**

Study	Tissue	Contrast	n_measured	mean log2FC	perm_p
OSD-120	root_tip	FLT_vs_GC_Dark	99	-0.239	0.0050
OSD-120	root_tip	FLT_vs_GC_Light	99	+0.059	0.4865
OSD-658	whole_seedling	GCR_40cGy_vs_NoRad	98	-0.245	0.0010
OSD-658	whole_seedling	GCR_80cGy_vs_NoRad	98	-0.109	0.0310
OSD-678	leaf	FLT_vs_GC_Dark	100	-0.462	0.4945
OSD-678	leaf	FLT_vs_GC_Light	100	-0.695	0.0010

Procambium_root (Han, n_panel=67) — Z = -3.50, BH-FDR = 9.4e-04, sign concordance = 5/6 **[SIGNIFICANT]**

Study	Tissue	Contrast	n_measured	mean log2FC	perm_p
OSD-120	root_tip	FLT_vs_GC_Dark	67	-0.438	0.0010
OSD-120	root_tip	FLT_vs_GC_Light	67	-0.446	0.0110
OSD-658	whole_seedling	GCR_40cGy_vs_NoRad	66	-0.137	0.1149
OSD-658	whole_seedling	GCR_80cGy_vs_NoRad	66	-0.125	0.0450
OSD-678	leaf	FLT_vs_GC_Dark	67	-0.302	0.9940
OSD-678	leaf	FLT_vs_GC_Light	67	+0.369	0.3946

Phloem_root (Han, n_panel=100) — Z = -2.98, BH-FDR = 0.005, sign concordance = 5/6 **[SIGNIFICANT]**

Study	Tissue	Contrast	n_measured	mean log2FC	perm_p
OSD-120	root_tip	FLT_vs_GC_Dark	99	-0.486	0.0010
OSD-120	root_tip	FLT_vs_GC_Light	99	-0.037	0.8881
OSD-658	whole_seedling	GCR_40cGy_vs_NoRad	98	-0.336	0.0010
OSD-658	whole_seedling	GCR_80cGy_vs_NoRad	98	-0.097	0.0509
OSD-678	leaf	FLT_vs_GC_Dark	99	-0.128	0.3586
OSD-678	leaf	FLT_vs_GC_Light	99	+0.795	0.0230

Liew_provasculature (Liew, n_panel=100) — Z = -2.20, BH-FDR = 0.039, sign concordance = 4/6 **[SIGNIFICANT]**

Study	Tissue	Contrast	n_measured	mean log2FC	perm_p
OSD-120	root_tip	FLT_vs_GC_Dark	98	-0.130	0.0350
OSD-120	root_tip	FLT_vs_GC_Light	98	-0.107	0.5165
OSD-658	whole_seedling	GCR_40cGy_vs_NoRad	97	-0.303	0.0010
OSD-658	whole_seedling	GCR_80cGy_vs_NoRad	97	-0.116	0.0230
OSD-678	leaf	FLT_vs_GC_Dark	98	+0.077	0.0410
OSD-678	leaf	FLT_vs_GC_Light	98	+0.372	0.3886

E.hypocotyl_epidermis_shoot (Han, n_panel=100) — Z = -1.11, BH-FDR = 0.335, sign concordance = 4/6 **[n.s.]**

Study	Tissue	Contrast	n_measured	mean log2FC	perm_p
OSD-120	root_tip	FLT_vs_GC_Dark	89	+0.666	0.0030
OSD-120	root_tip	FLT_vs_GC_Light	89	-0.007	0.9161
OSD-658	whole_seedling	GCR_40cGy_vs_NoRad	98	-0.083	0.4545
OSD-658	whole_seedling	GCR_80cGy_vs_NoRad	98	-0.090	0.0629
OSD-678	leaf	FLT_vs_GC_Dark	99	-0.913	0.0030
OSD-678	leaf	FLT_vs_GC_Light	99	+0.195	0.8521

hypocotyl_axis_composite (composite, n_panel=160) — Z = -0.51, BH-FDR = 0.681, sign concordance = 4/6 [n.s.]

Study	Tissue	Contrast	n_measured	mean log2FC	perm_p
OSD-120	root_tip	FLT_vs_GC_Dark	148	+0.640	0.0010
OSD-120	root_tip	FLT_vs_GC_Light	148	-0.254	0.0410
OSD-658	whole_seedling	GCR_40cGy_vs_NoRad	159	-0.056	0.8801
OSD-658	whole_seedling	GCR_80cGy_vs_NoRad	159	-0.075	0.0729
OSD-678	leaf	FLT_vs_GC_Dark	160	-0.680	0.0170
OSD-678	leaf	FLT_vs_GC_Light	160	+0.522	0.0569

Cortex_hypocotyl_shoot (Han, n_panel=67) — Z = -0.07, BH-FDR = 0.941, sign concordance = 4/6 [n.s.]

Study	Tissue	Contrast	n_measured	mean log2FC	perm_p
OSD-120	root_tip	FLT_vs_GC_Dark	65	+0.569	0.0290
OSD-120	root_tip	FLT_vs_GC_Light	65	-0.571	0.0010
OSD-658	whole_seedling	GCR_40cGy_vs_NoRad	67	-0.017	0.4306
OSD-658	whole_seedling	GCR_80cGy_vs_NoRad	67	-0.052	0.4266
OSD-678	leaf	FLT_vs_GC_Dark	67	-0.375	0.7522
OSD-678	leaf	FLT_vs_GC_Light	67	+1.094	0.0050